

Project Report

Bitcoin (BTC/USDT) Regime-Adaptive Trading System

With Research Pipeline, Machine Learning Filters, and TradingView Deployment

Field	Details
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Project type	Final-year / research-oriented trading system
Version	2.7 (July 2026)
Market / TF	BTCUSDT, 15-minute bars (+ 1h / daily context)

Abstract

This project designs, implements, and evaluates a **regime-adaptive Bitcoin trading system**. Market state is classified as trend, range, or volatility shock using ADX and ATR. Two playbooks fire accordingly: **trend pullback (A)** and **mean-reversion fade (B)**. The system enforces **one open position at a time**, asymmetric risk/reward (stop 1 ATR, take-profit 3 ATR on trend trades), and optional research filters drawn from academic literature (time-series momentum, VPIN / volume clock, walk-forward ML).

A **Python backtest engine** validates rules on Binance-style 15m history (2017–2026). A **TradingView Pine Script v6 indicator** (BTC Regime Gate v2.7) deploys the same logic for charting and alerts. Results are reported honestly: the best stable configuration (short_only) achieves roughly **+1%** on 2025 starting from \$100,000 while buy-and-hold was negative, and about **+\$1,000** summed across several calendar years—**not** a guaranteed 15% system. Experiments that chased higher returns (breakouts, ML spam entries) destroyed capital and were rejected.

1. Introduction

1.1 Motivation

Bitcoin alternates between strong trends and choppy ranges. Naive indicators either overtrade or enter late. Many retail systems also invert risk/reward (wide stop, tight target), which requires unrealistically high win rates after fees.

1.2 Objectives

1. Detect regime (trend / range / shock) with robust, explainable features.
2. Trade **one playbook per regime** with clear entry and exit rules.
3. Allow **only one long or one short** at a time (no signal spam).
4. Backtest with fees, slippage, and year-by-year evaluation.
5. Optionally use ML / microstructure filters without replacing the rule core.
6. Deploy the same logic on TradingView for live monitoring.
7. Document failures as carefully as successes (scientific honesty).

1.3 Scope

- **In scope:** Spot-style BTCUSDT OHLCV, 15m research, Pine indicator, offline backtests.
- **Out of scope:** Guaranteed returns, live brokerage execution, tick-level HFT, options VRP portfolios.

2. Literature Survey (Selected)

Papers were triaged into **KEEP** (used in design) and **NOT TAKEN** (theory / wrong market / too heavy for this stack).

2.1 KEEP (implemented or guiding design)

#	Paper	Use in this project
1	Easley, López de Prado, O'Hara — <i>The Volume Clock</i>	Volume-time thinking; volume buckets
2	Easley, López de Prado, O'Hara — <i>Flow Toxicity / VPIN</i>	Optional toxicity filter (<code>--vpin</code>)
3	Bailey & López de Prado — <i>Deflated Sharpe Ratio</i>	Avoid fake performance after many trials
4	Bailey et al. — <i>Probability of Backtest Overfitting</i>	Holdout alone is not enough
5	López de Prado — <i>Advances in Financial Machine Learning</i>	Triple-barrier labels, walk-forward, meta-label path
6	Carr & López de Prado — <i>Optimal Trading Rules without Backtesting</i>	Caution on over-tuning stops via search
7	Moskowitz, Ooi, Pedersen — <i>Time Series Momentum</i>	Side filter: trade with ~60-day return sign
8	Baz, Harvey et al. — <i>Dissecting carry/momentum/value</i>	Prefer time-series (single asset) over cross-section
9	Quant note — <i>Hyperparameters are trading opinions</i>	Treat windows/thresholds as risk beliefs

2.2 NOT TAKEN (examples)

CAPM (Sharpe), Fama–French cross-section, Markowitz portfolio theory, Kyle / Glosten–Milgrom (theory only), Bitcoin whitepaper (protocol), FinRL / deep RL libraries (overfit risk, deferred), Polymarket studies (wrong market), HFT queue/latency notes (not tradable here).

3. System Architecture

3.1 Two-layer design

Layer	Component	Role
A — Research	Python (<code>backtest_regime_gate.py</code> , <code>train_btc_patterns.py</code> , <code>vpin_volume.py</code>)	Features, backtest, ML, yearly stats
B — Deployment	Pine Script v6 (<code>BTC_Regime_Gate_v27.pine</code>)	Chart markers, position state, alerts

3.2 Decision flow

```
15m OHLCV bar
→ Volatility shock? → BLOCK
→ Regime: TREND / RANGE / NEUTRAL (ADX primary)
→ TS momentum side filter (optional, default ON)
→ Playbook A (trend pullback) or B (VWAP fade)
→ If FLAT + cooldown → open ONE position
→ Manage SL / TP / trail / time stop
→ EXIT (close long) or COVER (close short) → FLAT
```

3.3 Position discipline (critical)

Side	Open	Close
Long	BUY	EXIT (sell)
Short	SELL	COVER (buy back)

No second BUY while long; no second SELL while short. New entry only after flat + cooldown.

4. Methodology

4.1 Data

- Source: Binance-style BTCUSDT **15-minute** candles (`BTCUSDT_15m_All.csv`).
- Span used: approximately **2017-08 → 2026-06** (~310k bars).
- Resampling: 1h and 1D for higher-timeframe bias and TS momentum.

4.2 Regime detection

- **Trend:** $ADX \geq 30$ and no ATR shock.
- **Range:** $ADX \leq 18$ and no shock.
- **Shock:** $ATR / ATR_MA \geq 2.5 \rightarrow$ no new trades.
- HTF bias: 1h EMA50 vs EMA200 (lagged).
- Daily bias (Auto mode): price vs daily EMA50 / EMA200.

4.3 Playbook A — Trend pullback

- Structure: EMA21 vs EMA50 separation.
- Entry: pullback to EMA + reclaim candle + RVOL + HTF align.
- Exit: SL 1 ATR, TP 3 ATR, breakeven + trail, max hold ~96 bars.

4.4 Playbook B — Mean reversion

- Range regime + extreme rolling VWAP z-score + RSI extreme + reversal candle.
- Exit: tighter SL/TP and/or z-score mean return, short max hold.

4.5 Playbook C — Breakout (disabled)

Tested; produced large losses (~–15% in stress tests). **Kept OFF.**

4.6 Research add-ons

Module	Default	Role
Time-series momentum (~60d)	ON	Long only if momentum up; short if down
VPIN (bar-level proxy)	OFF (<code>--vpin</code>)	Stand aside when flow looks toxic
LightGBM pattern model	OFF (<code>--ml</code>)	Optional soft gate; walk-forward trained
Pure ML entries	Rejected	Overtrading; large losses OOS

4.7 Risk and costs (backtest)

Parameter	Value
Starting capital (per run)	\$100,000
Risk per trade	2% of equity

Max position fraction	25%
Slippage	~9 bps round-trip style
Fee	~4 bps one way
Trade modes	short_only, long_only, auto, both

5. Implementation

5.1 Repository layout (main files)

Path	Description
pine/BTC_Regime_Gate_v27.pine	Live indicator (one-position FSM)
python/backtest_regime_gate.py	Main backtest + yearly runner
python/train_btc_patterns.py	Feature build + LightGBM walk-forward
python/vpin_volume.py	Volume buckets + VPIN proxy
python/backtest_ml_2025.py	Honest OOS ML strategy test
python/models_btc15m/	Models, importance, yearly CSVs
STRATEGY.md	Locked strategy notes
docs/TECHNICAL_REPORT_v2.md	Earlier detailed design notes

5.2 How to run backtests

```
# Single year
python3 python/backtest_regime_gate.py \
  --data /path/to/BTCUSDT_15m_All.csv \
  --mode short_only --year 2025

# Multiple years (fresh $100k each)
python3 python/backtest_regime_gate.py \
  --data /path/to/BTCUSDT_15m_All.csv \
  --mode short_only \
  --years 2019,2020,2021,2022,2024,2025
```

5.3 TradingView deployment

- 1. Open Pine Editor → paste `BTC_Regime_Gate_v27.pine` → Add to chart (BTC 15m).
- 2. Remove older v2.1 / v2.3 indicators.
- 3. Verify table **Pos** = FLAT / LONG / SHORT and markers BUY→EXIT or SELL→COVER.

6. Experiments and Results

All figures below use **\$100,000** starting capital, fees/slippage on, breakout C off, unless noted.

6.1 Year-by-year — short_only (plain rules)

Year	Profit (approx.)	Return	Trades	Buy & hold (approx.)
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2019	−\$417	−0.42%	42	+95%
2020	+\$1,460	+1.46%	35	+303%
2021	+\$404	+0.40%	28	+61%
2022	−\$1,369	−1.37%	53	−64%
2024	−\$182	−0.18%	24	+120%
2025	+\$1,122	+1.12%	31	−6%
Sum	~\$+1,018			

Interpretation: edge is **small but real vs bad years for buy-and-hold (e.g. 2025)**; it does **not** capture full bull-market upside (2019–2021, 2024).

6.2 With TS momentum (v2.7)

On 2025 `short_only`, a smoke test after adding TS momentum showed about **+1.36%** with fewer trades (19)—directionally consistent with filtering against the trend sign.

6.3 Failed / weaker variants (documented)

Variant	Outcome	Decision
Breakout playbook C	Large loss (~−15% class)	Disabled
Pure ML probability entries	Catastrophic OOS (~−50% class)	Rejected
Aggressive VPIN (blocked most bars)	Almost no trades	Softened; opt-in only
Auto mode (longs + shorts)	Often worse than <code>short_only</code>	Prefer <code>short_only</code> / careful Auto

6.4 Machine learning summary

- ~116 engineered features; LightGBM walk-forward by year.
- OOS AUC ~0.51–0.55 (weak discrimination).
- Best use: **optional filter**, not primary signal generator.
- Model trained including 2025 is **not** a valid 2025 test; pre-2025 model used for honest checks.

7. Discussion

7.1 Why not 15%?

Academic and practitioner papers (Deflated Sharpe, backtest overfitting) warn that scanning many rules invents lucky history. On fee-aware BTC 15m, a **~1% annual edge with low drawdown** can still beat buy-and-hold in a down year, but is far from a marketing “15% printer.” Higher returns would need a different edge (e.g. strong multi-year trend capture on higher timeframes, or leverage—which multiplies risk).

7.2 Why one position?

Multiple stacked SELLS without COVER inflate risk and confuse evaluation. One ticket matches risk budgeting (2% per trade) and makes the Pine chart auditable.

7.3 Engineering contribution

The project is not only an indicator: it is a **full research loop**—data hygiene, regime rules, ablation of failed ideas, optional ML/VPIN, and chart deployment with matching position state.

8. Conclusion

This work delivered a **complete, honest BTC regime-gate trading project**:

1. Rule-based playbooks A/B with ADX regimes and HTF gates.
2. Strict **one-position** lifecycle (BUY/EXIT and SELL/COVER).
3. Research-backed filters (TS momentum; optional VPIN/ML).
4. Python backtests across multiple years with costs.
5. TradingView v2.7 indicator for live monitoring.
6. Clear rejection of overfitting paths that looked profitable only in hindsight.

Final recommended configuration for evaluation: `short_only` , SL 1 ATR / TP 3 ATR, 2% risk, TS momentum ON, breakout C OFF, VPIN/ML optional.

9. Future work

1. Meta-labeling (AFML): rules propose trade → ML only accepts/rejects.
 2. Report Deflated Sharpe with explicit trial count.
 3. Higher-timeframe (4h/1d) trend module for bull years.
 4. Tick-level VPIN if trade data becomes available.
 5. Paper trading / exchange API (out of current academic scope).
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10. References (selected)

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Appendix A — Glossary

Term	Meaning
ADX	Average Directional Index (trend strength)
ATR	Average True Range (volatility / stop sizing)
RVOL	Volume vs its recent average
VPIN	Volume-Synchronized Probability of Informed Trading
TS momentum	Sign of an asset's own past return
OOS	Out-of-sample (data not used to fit rules/model)
COVER	Buy back to close a short

Appendix B — Declaration

The numerical results in this report are from historical simulation. Past performance does not guarantee future results. This work is for **academic / educational** purposes and is not investment advice.

End of Report